```python
# ========================================
# Car Price Prediction using Linear Regression
# Google Colab Compatible
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# Step 1: Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Step 2: Create Sample Dataset
data = {
    'Year': [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2014, 2013, 2022],
    'Kilometers_Driven': [50000, 40000, 35000, 30000, 25000, 20000, 15000, 60000, 70000, 10000],
    'Fuel_Type': ['Petrol', 'Diesel', 'Petrol', 'Diesel', 'Petrol',
                  'Diesel', 'Petrol', 'Diesel', 'Petrol', 'Petrol'],
    'Price': [450000, 550000, 600000, 700000, 750000,
              850000, 900000, 400000, 350000, 1000000]
}

df = pd.DataFrame(data)

print("Dataset")
print(df)

# Step 3: Convert Categorical Data to Numerical
df = pd.get_dummies(df, columns=['Fuel_Type'], drop_first=True)

# Step 4: Features and Target
X = df.drop('Price', axis=1)
y = df['Price']

# Step 5: Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Step 6: Train Model
model = LinearRegression()
model.fit(X_train, y_train)

# Step 7: Prediction
y_pred = model.predict(X_test)

# Step 8: Evaluation
print("\nMean Absolute Error:", mean_absolute_error(y_test, y_pred))
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

# Step 9: Predict New Car Price
# Year, Kilometers_Driven, Fuel_Type_Diesel
new_car = pd.DataFrame({
    'Year': [2020],
    'Kilometers_Driven': [22000],
    'Fuel_Type_Petrol': [0]
})

predicted_price = model.predict(new_car)

print("\nPredicted Car Price: ₹{:,.0f}".format(predicted_price[0]))

# Step 10: Visualization
plt.figure(figsize=(6,5))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted Car Prices")
plt.grid(True)
plt.show()
```

```
Dataset
   Year  Kilometers_Driven Fuel_Type    Price
0  2015              50000    Petrol   450000
1  2016              40000    Diesel   550000
2  2017              35000    Petrol   600000
3  2018              30000    Diesel   700000
4  2019              25000    Petrol   750000
5  2020              20000    Diesel   850000
6  2021              15000    Petrol   900000
7  2014              60000    Diesel   400000
8  2013              70000    Petrol   350000
9  2022              10000    Petrol  1000000

Mean Absolute Error: 25169.4094838053
Mean Squared Error: 864492089.9479102
R2 Score: 0.913550791005209

Predicted Car Price: ₹856,467
```